Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3471-WE01

Title: (BEng/MEng)

Materials 3

Time Allowed:	2 hours
Additional Material provided:	Yes - Formula Sheet included as part of exam paper.
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

a) Determine the Miller indices of the planes in Figure Q1a. **[25%]**

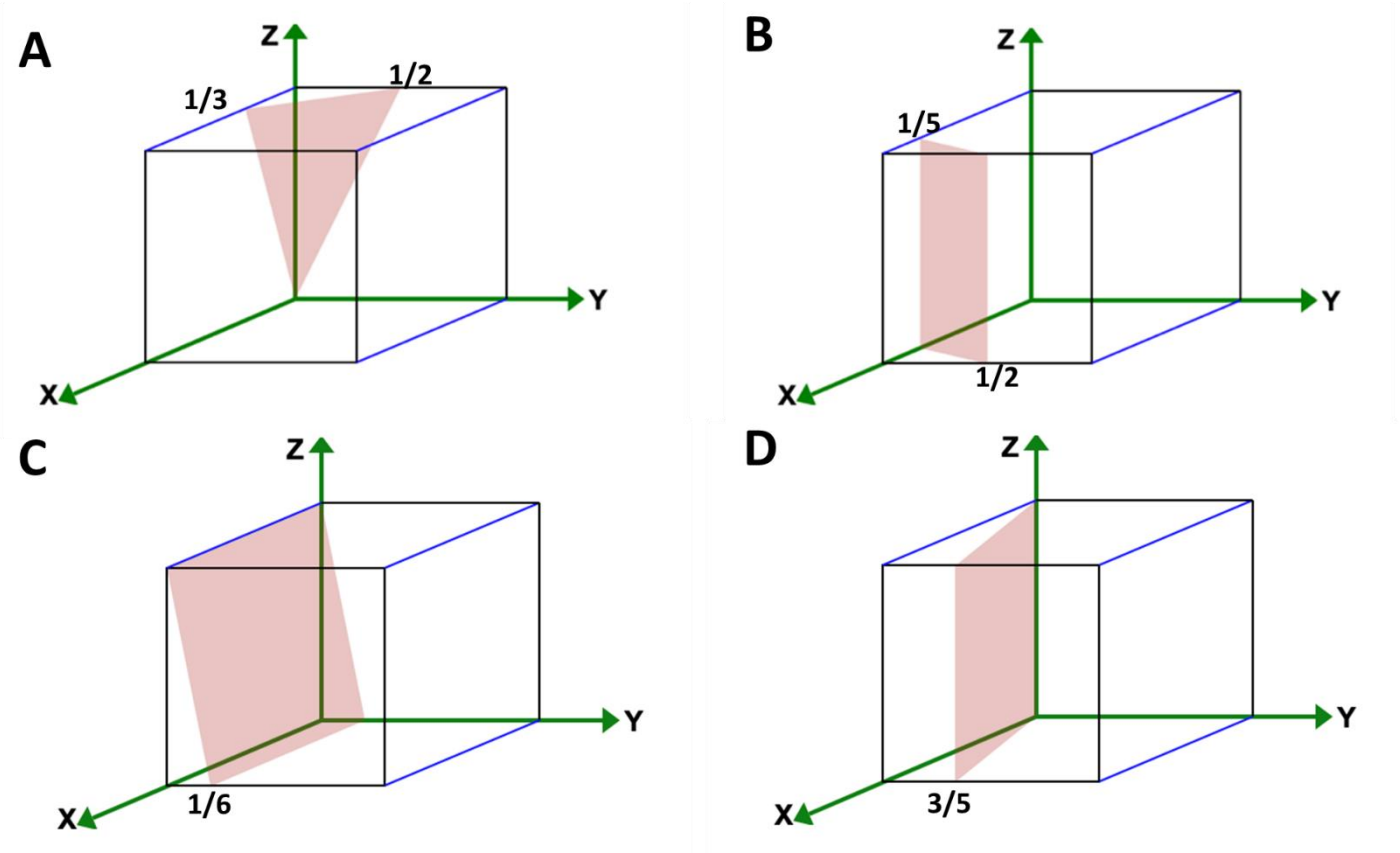


Figure Q1a: Crystallographic planes for Q1a

b) Figure Q1b shows the results of an X-ray diffraction (XRD) experiment. The materials used in the experiment is a mixture of powders containing various elements and compounds (all crystallized in the Cubic system). Utilizing the information in Figure Q1b and Bragg's Law, determine the number of materials present in the experiment. Note that X-rays with a wavelength of 0.15418 nm are employed. **[75%]**

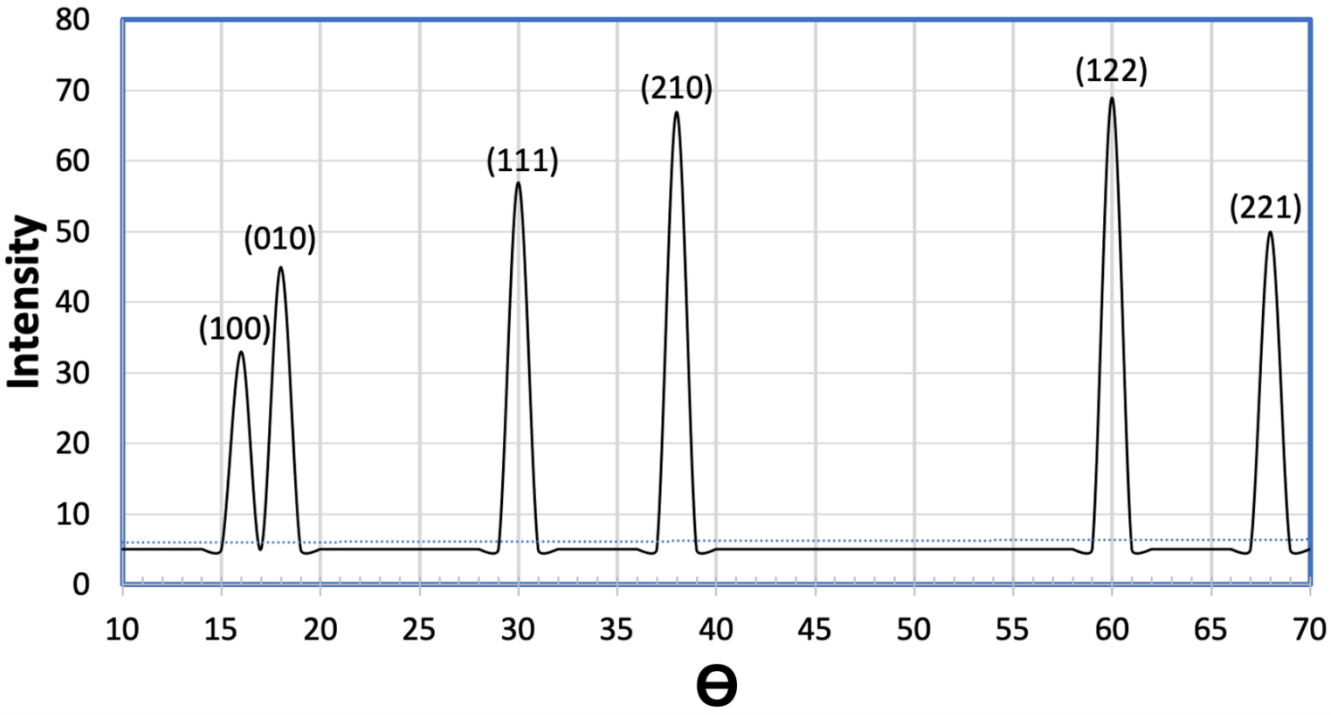


Figure Q1b: XRD pattern for Q1b

continued

Question 2

a) Thin sheets of low carbon steel in different conditions—annealed, cold-worked, and recrystallized—are immersed in an electrolyte. Which of these is more prone to corrosion, and which is less likely to corrode? Provide an explanation. **[30%]**

b) The binary phase diagram of Ni-Ti is shown in Figure Q2b. How many three phase reactions can you identify in this phase diagram? Write and name 3 reactions and identify the composition and ratio of the phases at “just below” the reaction temperature. **[50%]**

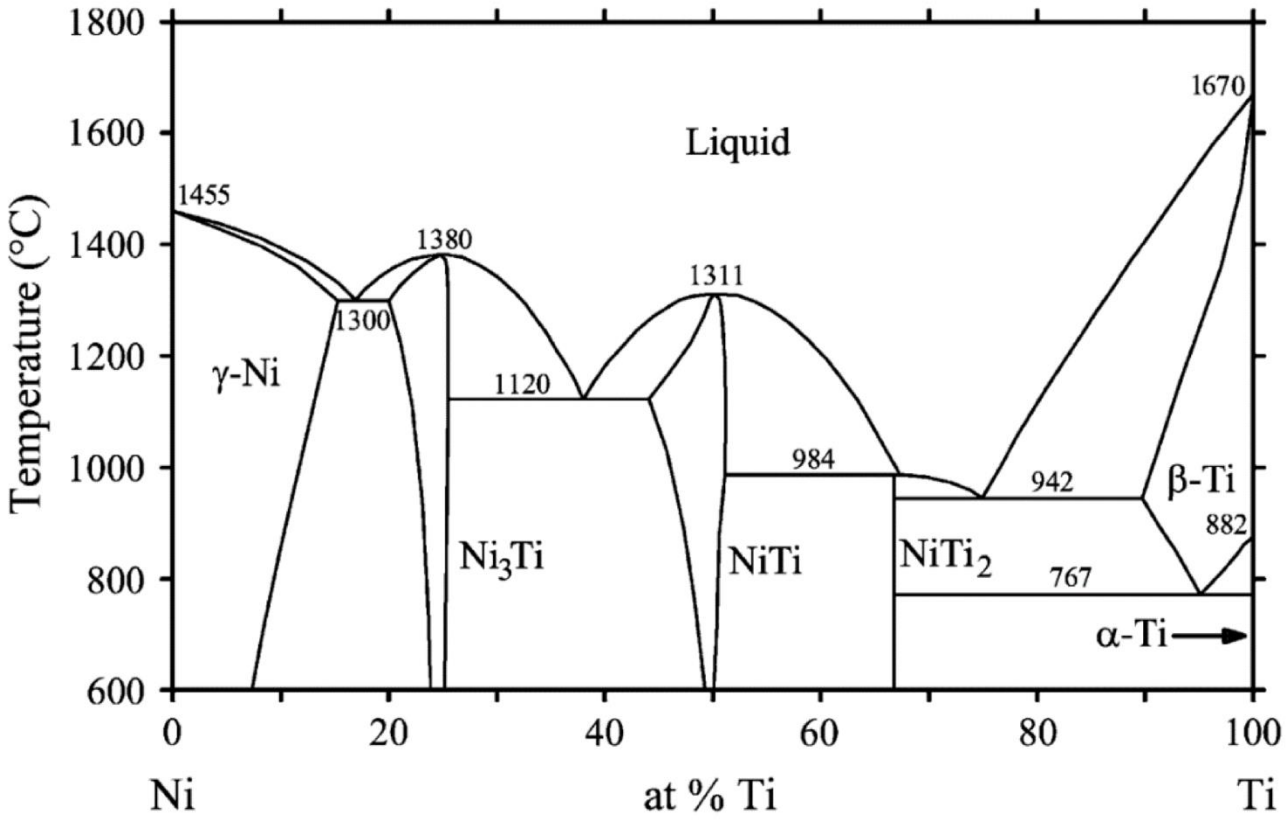


Figure Q2b: Phase Diagram for Q2b

continued

c) Evaluating a polycrystalline polymer sample (refer to Figure Q2c) reveals the existence of six chain lengths, each with a specific number of chains. Calculate (i) the weight-average molecular weight, and (ii) the number-average molecular weight. **[20%]**

Number of Chains	Mean Molecular Weight of Chains (g/mol)
10,000	3,000
18,000	6,000
17,000	9,000
15,000	12,000
9,000	15,000
4,000	18,000

Figure Q2c: Polymer data for Q2c

Question 3

a) How does “Hot working” and “cold working” compare from a sustainability point of view? Provide a qualitative life cycle assessment (LCA) of the two processes and present your assumptions. **[40%]**

b) As an engineer, you oversee developing and optimizing a fabrication process for manufacturing a battery intended for a portable electronic device. What factors should be considered during the design phase to promote sustainability and minimize the carbon footprint of the production of the product? Discuss your suggestion from materials selection, design, fabrication to end of life. **[60%]**

continued

Question 4

With environmental concerns, the use of electricity instead of fossil fuels has never been greater. A bus company is considering running a fleet of trolleybuses using a set of electrical cables that is extended aerially throughout the entire city network (see sketch, Figure Q4). The R&D team within the company met several times to consider the selection of suitable materials for the electrical cable.

a) Meeting 1: The leader of the team announced that the initial specification was to only minimise the cable resistance (R) for a given cross-sectional area (A) and proposed 5 materials listed in Table Q4. Using $R = \frac{L\rho_e}{A}$, state the best performing material in Table Q4? **[10%]**

b) Meeting 2: The leader of the team shared that there are budget constraints, the cost needs to be considered and as such the new specification was to minimise cable resistance for a given raw material cost budget. Derive an appropriate Material Index and identify the best performing material in Table Q4. **[40%]**

c) Meeting 3: A discussion took place to consider the combination of the power loss in the cable (I^2R) and the energy consumption associated with the pole's weight contribution to the trolleybus mass. Reducing the total mass of the trolleybus would save in energy consumption; every gram (g) saved on the mass of the trolleybus would save 0.005 Watts, and the overall root mean square (RMS) of the current in the cable is I = 65 Amps.

Considering the combined power loss, use the data in Table Q4 to:

i. Derive an expression for the optimal cable cross sectional area for a given set of material properties. **[20%]**

ii. Derive a material index to minimise the combined energy loss associated with the cable resistance and the pole's mass (which can be assumed to be made of the same material as the cable). **[20%]**

iii. Select the best performing material. **[10%]**

continued

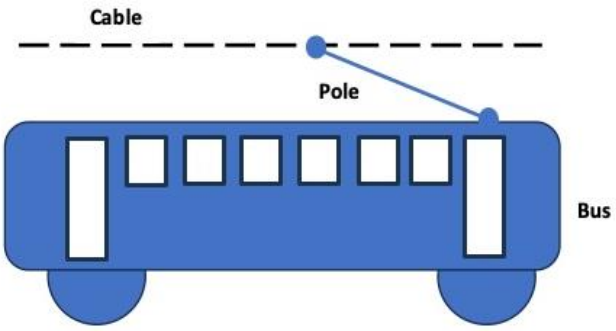


Figure Q4 – Trolleybus sketch – not to scale.

Material	**Density** ρ **(kg/m^3)**	**Cost per unit mass** C_M **(£/kg)**	**Yield Stress** σ_f **(MPa)**	**Electrical Resistivity** ρ_e **(Ω m)**
Aluminium Alloy (6201-T81)	2700	1.54	330	3.16E-08
Copper Alloy	8960	5.22	410	1.68E-08
Magnesium Alloy	1850	2.31	220	1.03E-07
Titanium Alloy (6Al-4V)	4460	16.45	900	1.71E-06
Carbon Fibre	1500	27.30	1200	1.43E-06

Table Q4 – Material Properties

continued

Question 5

a) Miller Material Services Ltd is a metal powder manufacturer specialising in converting solid blocks of brass into metallic powders of spherical shape using gas atomization. Describe your understanding of gas atomization. **[40%]**

b) Heat treatment involves various heating and cooling procedures performed to affect microstructural changes in a material, which in turn affects its mechanical properties.

i. Describe the main steps in the heat treatment process for metals. **[30%]**

ii. George Taylor runs a small engineering firm producing cutting tools used in lathe machines. The cutting tools are made from high-speed steel. Explain why a heat treatment needs to be applied to the cutting tools and which heat treatment process should be used. Justify your choice in your own words. **[30%]**

Formula sheet

Bragg's Law:

$$sin\ \theta = \frac{\lambda}{2d_{hkl}}$$

Interplanar spacing:

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Concentration of vacancies:

$$n_v = n \exp(-Q_v/RT)$$

ASTM grain size number:

$$N = 2^{n-1}$$

Percent cold work:

$$Percent\ of\ cold\ work = \left[\frac{A_0 - A_f}{A_0}\right] \times 100$$

Hall-Petch Equation:

$$\sigma_y = \sigma_0 + kd^{-1/2}$$

Diffusion coefficient dependency on Temperature

$$D = D_0 \exp(\frac{-Q}{RT})$$

Stress relaxation:

$$\sigma = \sigma_0 \exp(-t/\lambda)$$

Relaxation time:

$$\lambda = \lambda_0 \exp(Q/RT)$$

The weight average molecular weight:

$$\overline{M_w} = \sum f_i M_i$$

continued

The number average molecular weight:

$$\overline{M_n} = \sum x_i M_i$$

Screen mesh for sorting particle sizes

$$PS = \frac{1}{MC} - t_w$$

Where PS = particle size, MC = mesh count, openings per linear mm and t_w = wire thickness of screen mesh.

Surface area (sphere)

$$A = \pi D^2$$

Where D = diameter of the spherical particle

Surface volume (sphere)

$$V = \frac{\pi D^3}{6}$$

Where D = diameter of the spherical particle

Area-to-volume ratio for a sphere

$$\frac{A}{V} = \frac{6}{D}$$

Where A = surface area, V = surface volume, D = diameter of the spherical particle

Area-to-volume ratio for any particle shape – spherical or non-spherical

$$\frac{A}{V} = \frac{K_S}{D} \text{ or } K_s = \frac{AD}{V}$$

Where K_s = shape factor; D = the diameter of a sphere of equivalent volume as the nonspherical particle.

continued

Power density

$$PD = \frac{P}{A}$$

where PD = power density, W/mm^2; P = power entering the surface, W; and A= surface area over which the energy is entering, mm^2.

Heat required to melt a given volume of metal

$$U_m = KT_m^2$$

where U_m = the unit energy for melting, J/mm^3; T_m = melting point of the metal on an absolute temperature scale, °K(°R); and K = constant whose value is 3.33 x 10^{-6} JK^{-2}mm^{-3}, when the Kelvin scale is used (and K = 1.467 x 10^{-5} JR^{-2}mm^{-3} for the Rankine temperature scale).

Heat energy available for welding:

$$H_w = f_1 f_2 H$$

where H_w = net heat available for welding, J, f_1 = heat transfer factor, f_2 = the melting factor, and H = the total heat generated by the welding process, J.

Energy balance equation

$$H_w = U_m V$$

Where H_w = net heat energy used by the welding operation, J; U_m = unit energy required to melt the metal, J/mm^3; and V = the volume of metal melted, mm^3.

Rate balance equation

$$R_{Hw} = U_m R_{WV}$$

Where R_{Hw} = rate of heat energy delivered to the operation for welding, J/s = W; and R_{WV} = volume rate of metal welded, mm^3/s.

$$R_{Hw} = f_1 f_2 R_H = U_m A_w v$$

here f_1 and f_2 are the heat transfer and melting factors; R_H = rate of input energy generated by the welding power source, W; A_w = weld cross-sectional area, mm^2; and v = the travel velocity of the welding operation, mm/s.

continued

Time to complete a single layer in rapid prototyping

$$T_i = \frac{A_i}{vD} + T_r$$

Where T_i= time to complete layer i; A_i = area of layer i; v = average scanning speed of the laser beam at the surface; D = diameter of the laser beam at the surface, and T_r = repositioning time between layers.

Build cycle time (stereolithography)

$$T_c = \sum_{i=1}^{nl} T_i$$

Where T_c = the STL build cycle time; and nl = the number of layers used to approximate the part.

END